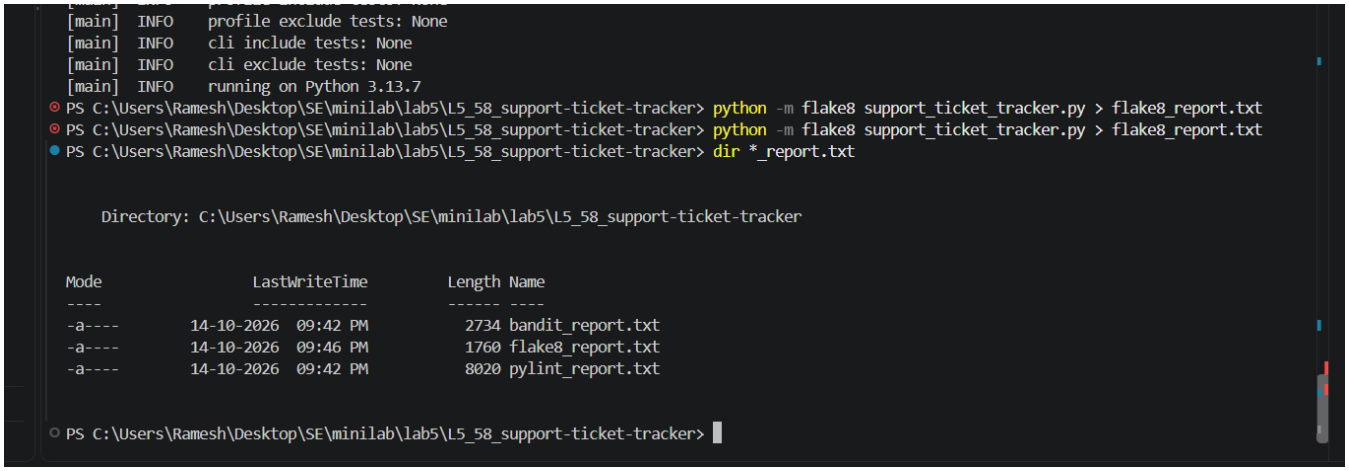
NAME:Shrungashree KR

Sec: H

SRN:PES2UG24CS497

SE -Lab5 report

Before Static Analysis — Report Generation



1. Pylint

- Before: Pylint reported several code-quality issues, including missing docstrings, unused variables, a mutable default argument, and long lines, with a score of 5.00/10.

```
PS C:\Users\Ramesh\Desktop\SE\minilab\lab5\L5_58_support-ticket-tracker> Get-Content .\pylint_report_before.txt
************* Module support_ticket_tracker
support_ticket_tracker.py:10:0: C0301: Line too long (101/100) (line-too-long)
support_ticket_tracker.py:14:0: C0301: Line too long (176/100) (line-too-long)
support_ticket_tracker.py:84:24: C0303: Trailing whitespace (trailing-whitespace)
support_ticket_tracker.py:1:0: C0114: Missing module docstring (missing-module-docstring)
support_ticket_tracker.py:10:0: C0116: Missing function or method docstring (missing-function-docstring)
support_ticket_tracker.py:10:0: W0102: Dangerous default value [] as argument (dangerous-default-value)
support_ticket_tracker.py:10:0: R0913: Too many arguments (6/5) (too-many-arguments)
support_ticket_tracker.py:10:0: R0917: Too many positional arguments (6/5) (too-many-positional-arguments)
support_ticket_tracker.py:16:21: C0209: Formatting a regular string which could be an f-string (consider-using-f-string)
support_ticket_tracker.py:20:0: C0116: Missing function or method docstring (missing-function-docstring)
support_ticket_tracker.py:24:4: W0702: No exception type(s) specified (bare-except)
support_ticket_tracker.py:20:36: W0613: Unused argument 'notify' (unused-argument)
support_ticket_tracker.py:29:0: C0116: Missing function or method docstring (missing-function-docstring)
support_ticket_tracker.py:34:0: C0116: Missing function or method docstring (missing-function-docstring)
support_ticket_tracker.py:37:11: R1714: Consider merging these comparisons with 'in' by using 'status not in (None, t['status'])'. Use a set
instead if elements are hashable. (consider-using-in)
support_ticket_tracker.py:37:11: C0121: Comparison 'status != None' should be 'status is not None' (singleton-comparison)
support_ticket_tracker.py:45:0: C0116: Missing function or method docstring (missing-function-docstring)
support_ticket_tracker.py:46:11: W1514: Using open without explicitly specifying an encoding (unspecified-encoding)
support_ticket_tracker.py:46:11: R1732: Consider using 'with' for resource-allocating operations (consider-using-with)
support_ticket_tracker.py:51:0: C0116: Missing function or method docstring (missing-function-docstring)
support_ticket_tracker.py:52:4: W0603: Using the global statement (global-statement)
support_ticket_tracker.py:53:11: W1514: Using open without explicitly specifying an encoding (unspecified-encoding)
```

- After: After fixing the issues, the Pylint score increased to 9.61/10, exceeding the required 9.5.

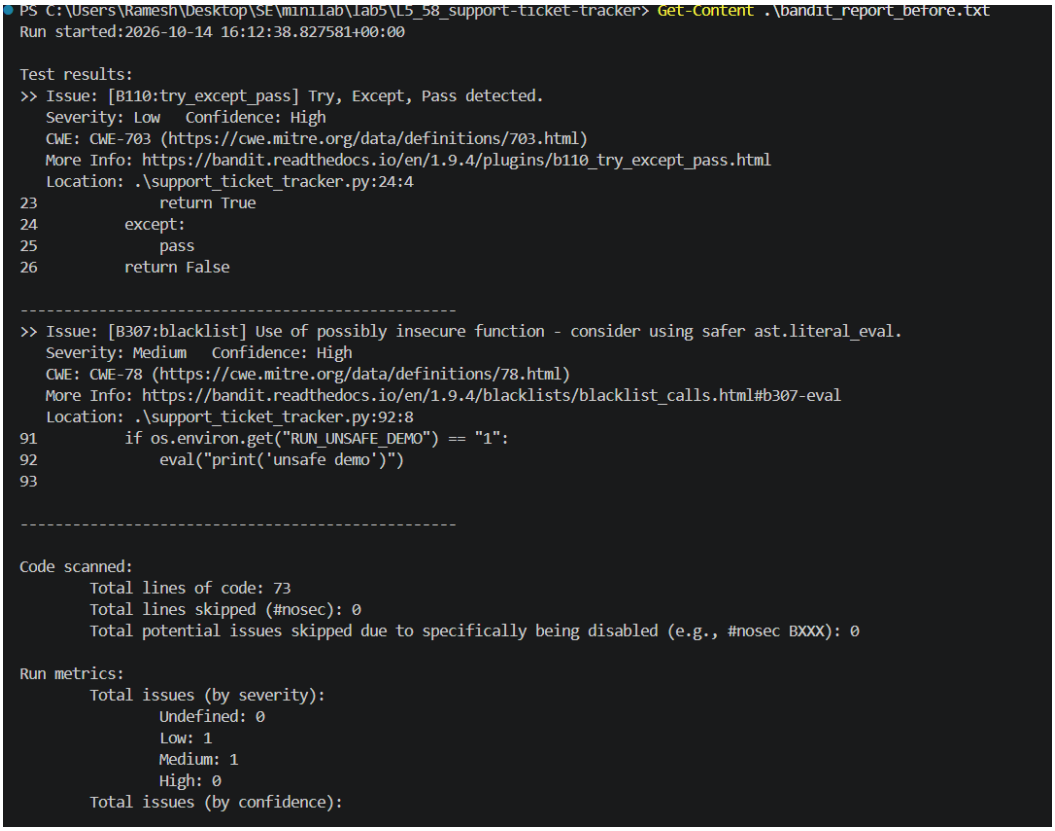
```
PS C:\Users\Ramesh\Documents\SE_LAB_PES2UG24CS497> python -m pylint .\lab5\L5_58_support-ticket-tracker\support_ticket_tracker.py
************* Module support_ticket_tracker
lab5\L5_58_support-ticket-tracker\support_ticket_tracker.py:11:0: R0913: Too many arguments (6/5) (too-many-arguments)
lab5\L5_58_support-ticket-tracker\support_ticket_tracker.py:11:0: R0917: Too many positional arguments (6/5) (too-many-positional-arguments)
lab5\L5_58_support-ticket-tracker\support_ticket_tracker.py:83:4: W0603: Using the global statement (global-statement)

-------------------------------------------------------------------
Your code has been rated at 9.61/10 (previous run: 5.00/10, +4.61)

PS C:\Users\Ramesh\Documents\SE_LAB_PES2UG24CS497>
```

2. Bandit

- Before: Bandit detected two security issues: broad exception handling and the use of eval(), which can execute unsafe code.



```
PS C:\Users\Ramesh\Desktop\SE\minilab\lab5\L5_58_support-ticket-tracker> Get-Content .\bandit_report_before.txt
Run started:2026-10-14 16:12:38.827581+00:00

Test results:
>> Issue: [B110:try_except_pass] Try, Except, Pass detected.
   Severity: Low   Confidence: High
   CWE: CWE-703 (https://cwe.mitre.org/data/definitions/703.html)
   More Info: https://bandit.readthedocs.io/en/1.9.4/plugins/b110_try_except_pass.html
   Location: .\support_ticket_tracker.py:24:4
23              return True
24          except:
25              pass
26          return False

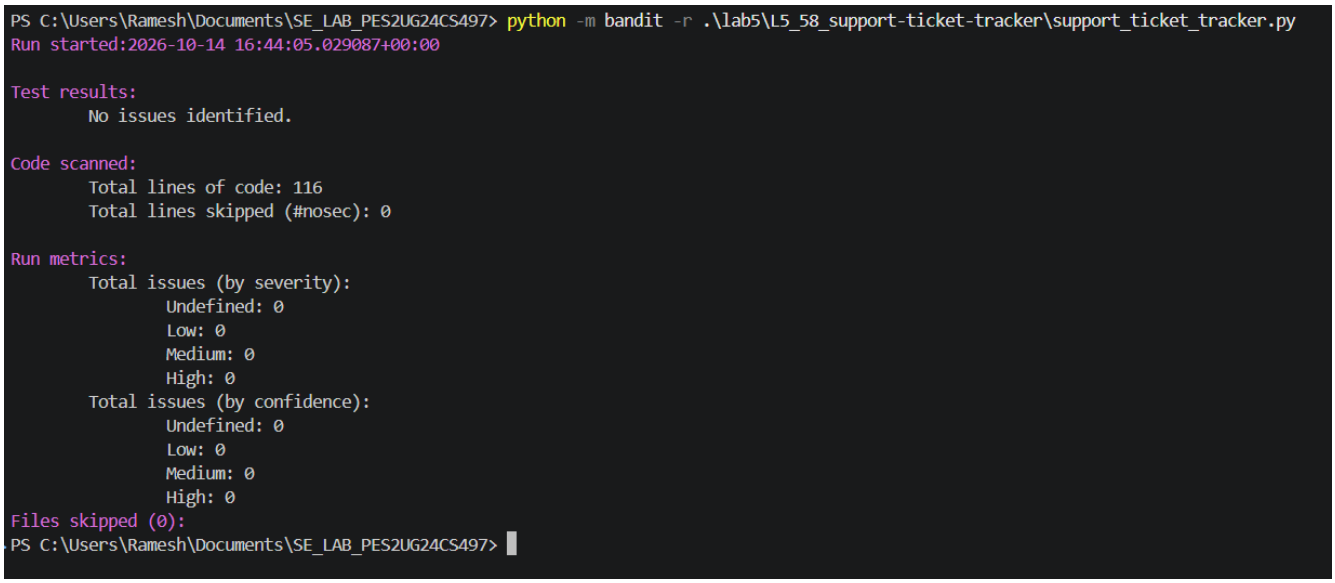
--------------------------------------------------
>> Issue: [B307:blacklist] Use of possibly insecure function - consider using safer ast.literal_eval.
   Severity: Medium   Confidence: High
   CWE: CWE-78 (https://cwe.mitre.org/data/definitions/78.html)
   More Info: https://bandit.readthedocs.io/en/1.9.4/blacklists/blacklist_calls.html#b307-eval
   Location: .\support_ticket_tracker.py:92:8
91          if os.environ.get("RUN_UNSAFE_DEMO") == "1":
92              eval("print('unsafe demo')")
93

--------------------------------------------------

Code scanned:
        Total lines of code: 73
        Total lines skipped (#nosec): 0
        Total potential issues skipped due to specifically being disabled (e.g., #nosec BXXX): 0

Run metrics:
        Total issues (by severity):
                Undefined: 0
                Low: 1
                Medium: 1
                High: 0
        Total issues (by confidence):
```

- After: After the fixes, Bandit reported “No issues identified”, with zero findings at all severity levels.



```
PS C:\Users\Ramesh\Documents\SE_LAB_PES2UG24CS497> python -m bandit -r .\lab5\L5_58_support-ticket-tracker\support_ticket_tracker.py
Run started:2026-10-14 16:44:05.029087+00:00

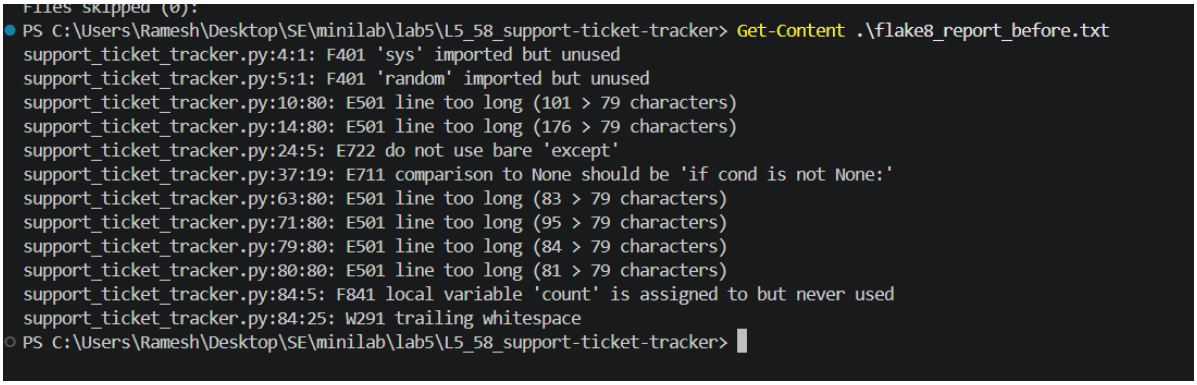
Test results:
        No issues identified.

Code scanned:
        Total lines of code: 116
        Total lines skipped (#nosec): 0

Run metrics:
        Total issues (by severity):
                Undefined: 0
                Low: 0
                Medium: 0
                High: 0
        Total issues (by confidence):
                Undefined: 0
                Low: 0
                Medium: 0
                High: 0
Files skipped (0):
PS C:\Users\Ramesh\Documents\SE_LAB_PES2UG24CS497>
```

3. Flake8

- Before: Flake8 reported multiple style violations, including unused imports, long lines, and trailing whitespace.



- After: After correcting the formatting issues, Flake8 produced no output, indicating that no style violations were found

